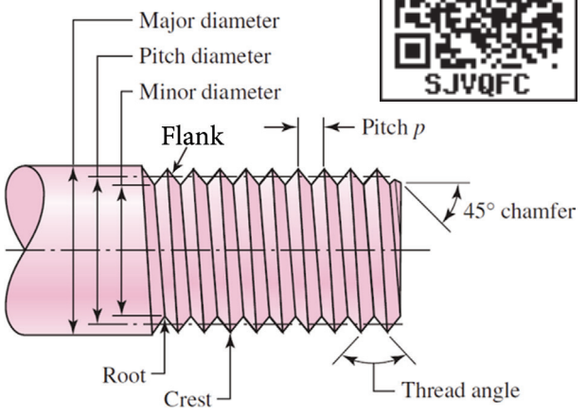
7.3 THREAD

Threads are first developed by sir Joseph Whitworth of England in the year 1841. Threads are one of the several methods of fastening two or more components temporarily. A thread is a helical groove cut on the cylindrical surface of a round rod or a hole.



Nomenclature of Thread

Nomenclature of Threads

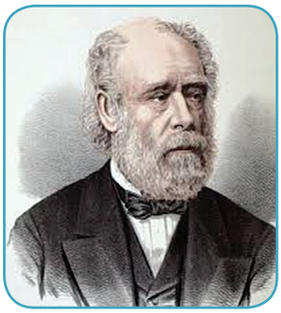
Major Diameter: This is the largest diameter of a screw thread, touching the crests of an external thread or the roots of an internal thread. A screw is specified by its major diameter.

Minor Diameter: This is the smallest diameter of a screw thread, touching the roots or core of an external thread(or) the crests of an internal threads.

Crest: It is the peak edge of the screw thread that connects adjacent the flanks of a thread at the top.

Root: It is the bottom edge of the thread that connects the adjacent flanks of a thread at the bottom.

DO YOU KNOW?



Joseph Whitworth was one of the greatest Victoria Mechanical Engineers. He is master of Screws.

Flank: The inclined flat surface of the thread which connects the crest and the root is called flank.

Depth of Thread: The depth of the thread is the distance between the crest and root which is measured right angle to the axis.

Pitch: Pitch is the distance measured parallel to the axis, between on one thread to the corresponding point on the adjacent screw threads.

TPI: Number of threads per inch in a screw is abbreviated as TPI.

Lead: It is defined as the distance which a screw moves axially in one complete rotation. Lead is equal to 1/TPI and is equal to pitch.

TYPES OF THREADS

Thread are classified into two types V-shape thread and square shape thread.

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